

Gait-Based Human Identification Using Deep Learning

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Abstract

This project proposes a deep learning-based gait recognition system for human identification using the CASIA-B dataset. The system identifies a person from walking style using silhouette image sequences instead of face or fingerprint biometrics. A 2D Convolutional Neural Network (CNN) is used to extract frame-level gait features, followed by Temporal Max Pooling for sequence-level feature aggregation. The final output is a 256-dimensional gait embedding used for person identification. Training is performed using Cross-Entropy Loss with label smoothing, dropout, learning rate scheduling, and early stopping to prevent overfitting. Evaluation is performed using cosine distance-based retrieval with Rank-1 and Rank-5 accuracy. The model achieves strong same-view and multi-view performance on unseen subjects under normal, bag, and coat walking conditions.

1 Introduction

Biometric identification is an important research area in computer vision and security systems. Traditional biometric systems mainly rely on face recognition, fingerprint recognition, iris scanning, and voice recognition. However, these methods often require close-range interaction or user cooperation.

Gait recognition is a behavioral biometric technique that identifies a person using their walking pattern. It can work from a long distance without physical contact, making it highly suitable for surveillance, public security, forensic investigation, and smart city monitoring.

The major challenge in gait recognition is maintaining recognition accuracy under different real-world conditions such as:

- View angle variation
- Clothing changes

- Carrying conditions (bags, backpacks)
- Occlusion and silhouette noise

This project focuses on solving gait-based human identification using deep learning on the CASIA-B benchmark dataset. Instead of using traditional Gait Energy Image (GEI), the proposed method uses sequence-based silhouette learning where multiple frames are processed individually and then aggregated using Temporal Max Pooling.

The main objectives are:

- To preprocess gait silhouette sequences efficiently
- To design a robust CNN-based gait feature extractor
- To generate discriminative 256-dimensional gait embeddings
- To evaluate performance using Rank-1 and Rank-5 accuracy
- To analyze same-view and multi-view gait recognition performance

2 Literature Review

Gait recognition has been an active research area in biometric identification because it allows recognition of individuals from a distance without physical contact or subject cooperation. Over the years, researchers have proposed several approaches ranging from traditional handcrafted methods to modern deep learning-based architectures.

2.1 Traditional Template-Based Methods

One of the earliest and most influential works in gait recognition was proposed by Han and Bhanu [?], called the **Gait Energy Image (GEI)**. In this method, multiple silhouette frames from a walking sequence are averaged into a single grayscale template called GEI.

The major advantages of GEI are:

- Simple implementation
- Low computational cost
- Fast matching

However, GEI has an important limitation:

- It loses temporal walking dynamics because multiple frames are compressed into a single static image

As a result, performance decreases under clothing variation, carrying conditions, and view-angle changes.

2.2 Model-Based Gait Recognition

Another category of gait recognition methods uses model-based approaches, where body joints, skeleton motion, and human pose estimation are extracted for recognition.

These methods focus on:

- joint trajectories
- limb movement
- skeleton-based walking patterns

Although these methods provide strong discriminative information, they require:

- high-resolution video
- accurate pose estimation
- high computational resources

Therefore, they are difficult to deploy in real-world surveillance environments.

2.3 Deep Learning-Based Methods

Recent research has shifted toward deep learning approaches for gait recognition.

One of the most important works is **GaitSet** proposed by Chao et al. [?]. Instead of converting gait into a single image like GEI, GaitSet treats gait as a **set of independent silhouette frames**. Each frame is processed individually using CNNs, followed by Set Pooling to generate a final gait representation.

The advantages of GaitSet include:

- preservation of temporal information
- better cross-view recognition
- improved robustness under clothing and carrying conditions

Another important work is **GaitPart**, which further improves performance by learning fine-grained body-part level gait features instead of using whole-body representation only.

These methods significantly improved state-of-the-art gait recognition performance on the CASIA-B dataset.

2.4 CASIA-B Dataset

The CASIA-B dataset proposed by Yu et al. [?] is one of the most widely used benchmark datasets for gait recognition research.

It contains:

- 124 subjects
- 11 view angles

- Normal walking (NM)
- Walking with bag (BG)
- Walking with coat (CL)

Most modern gait recognition papers use CASIA-B for performance evaluation and comparison.

2.5 Research Gap and Our Contribution

From the literature survey, it is observed that:

- GEI-based methods are simple but lose temporal walking information
- Model-based methods are accurate but computationally expensive
- Advanced methods like GaitSet and GaitPart achieve strong performance but are often complex and heavy for student-level implementation

Our work focuses on building a practical and efficient deep learning-based gait recognition system inspired by modern sequence-based learning methods.

The main difference between previous work and our proposed method is:

Table 1: Comparison of Existing Work and Proposed Method

Method	Existing Work	Our Proposed Work
Feature Type	Static GEI / hand-crafted features	Dynamic silhouette sequence learning
Temporal Information	Mostly lost	Preserved using Temporal Max Pooling
Model Complexity	Very simple or very heavy	Lightweight CNN-based architecture
Evaluation	Mostly same-view	Same-view + Multi-view evaluation
Identification Method	Classification-based	Retrieval using cosine distance
Robustness	Limited under BG/CL conditions	Improved performance under BG and CL

Our model uses:

- frame-wise CNN feature extraction
- Temporal Max Pooling
- 256-dimensional gait embedding
- CrossEntropy loss with label smoothing
- Rank-1 and Rank-5 retrieval evaluation

Thus, the proposed system is closer to modern research-level gait recognition systems while remaining computationally efficient, interpretable, and suitable for practical implementation.

3 Proposed Methodology

The complete system consists of four major stages:

1. Data preprocessing
2. Dataset preparation and splitting
3. Deep learning model training
4. Retrieval-based evaluation

3.1 Data Preprocessing

The raw CASIA-B dataset contains binary silhouette images of walking subjects under multiple conditions and view angles.

The preprocessing pipeline includes:

1. Read grayscale silhouette image
2. Detect all non-zero pixels
3. Generate bounding box around the walking subject
4. Remove unnecessary background regions
5. Resize cropped silhouette to 64×64 pixels
6. Store complete gait sequence as `.pkl` file

This improves computational efficiency and ensures consistent model input.

3.2 Dataset Split

CASIA-B originally provides standard train and test subject separation:

- Subjects 001–074 : Training split
- Subjects 075–124 : Testing split

However, deep learning requires a validation set for model selection and overfitting control. Therefore, we created an internal validation split from the training subjects. it is the sample based validation.

Final dataset split used in this project:

Table 2: Final Dataset Split		
Split	Subject IDs	Number of Subjects
Training and validation	001–074	74
Testing	075–124	50

Validation subjects are used for:

- validation loss calculation
- validation accuracy monitoring
- learning rate scheduling
- early stopping
- best model checkpoint saving

Testing subjects are never used during training.

3.3 Model Architecture

The proposed model uses a lightweight CNN followed by Temporal Max Pooling.

3.3.1 CNN Backbone

Each silhouette frame is processed independently using three convolutional blocks.

- Conv Layer 1: $1 \rightarrow 32$ channels
- Conv Layer 2: $32 \rightarrow 64$ channels
- Conv Layer 3: $64 \rightarrow 128$ channels
- Batch Normalization
- ReLU activation
- Max Pooling

After convolution, frame-level features are generated.

3.3.2 Temporal Max Pooling

For each gait sequence, 30 consecutive frames are used.

Temporal Max Pooling is applied across the frame dimension to extract the strongest gait features and make the model robust to missing or noisy frames.

This makes the system frame-invariant.

3.3.3 Fully Connected Embedding Layer

The pooled feature vector passes through fully connected layers to generate a final 256-dimensional embedding representing the gait signature of a person.

3.4 Training Strategy

Training includes the following optimization techniques:

- Cross-Entropy Loss with label smoothing = 0.1
- Dropout regularization = 0.5
- Adam optimizer
- ReduceLROnPlateau learning rate scheduler
- Early stopping (patience = 7)
- Best model checkpoint saving

An external classifier layer is used only during training for subject classification. During testing, only the embedding extractor is used.

4 Experimental Result

4.1 Dataset Used

We use the CASIA-B gait dataset, one of the most widely used benchmark datasets for gait recognition research.

It contains:

- 124 subjects
- 11 view angles
- Normal walking (NM)
- Walking with bag (BG)
- Walking with coat (CL)

The view angles are:

000°, 018°, 036°, 054°, 072°, 090°, 108°, 126°, 144°, 162°, 180°

4.2 Experimental Settings

Training settings:

Table 3: Training Configuration

Parameter	Value
Batch Size	16
Learning Rate	0.001
Epochs	50
Frame Count	30
Embedding Dimension	256
Loss Function	Cross-Entropy
Optimizer	Adam
Scheduler	ReduceLROnPlateau
Dropout	0.5
Early Stopping	Yes

Best validation accuracy achieved:

99.82% at Epoch 26

4.3 Same-View Experimental Results

Evaluation is performed as retrieval using cosine distance.

Gallery:

NM-01 at 090°

Probe:

NM-02, BG-01, CL-01 at 090°

Table 4: Same-View Retrieval Accuracy

Condition	Rank-1 Accuracy	Rank-5 Accuracy
NM-02	98.00%	100.00%
BG-01	86.00%	98.00%
CL-01	76.00%	90.00%
Average	86.67%	96.00%

The results show that clothing variation is the most difficult condition because coat-based walking changes body silhouette significantly.

4.4 Multi-View Experimental Results

Gallery:

NM-01 at 090°

Probe:

NM-02 at all view angles

Table 5: Multi-View Rank-1 and Rank-5 Accuracy

Angle	Rank-1	Rank-5
000°	24.00%	64.00%
018°	44.00%	76.00%
036°	54.00%	80.00%
054°	60.00%	84.00%
072°	90.00%	98.00%
090°	98.00%	100.00%
108°	96.00%	100.00%
126°	92.00%	100.00%
144°	76.00%	94.00%
162°	48.00%	80.00%
180°	26.00%	64.00%
Average	64.36%	85.45%

Accuracy varies across angles because gait recognition is highly view-dependent. Side-view angles provide maximum walking dynamics, while frontal and rear views reduce visible gait information. This is known as the **view variation problem**.

4.5 Time Complexity

The proposed model uses lightweight 2D CNNs instead of heavy 3D CNNs, reducing computational complexity significantly.

Advantages:

- Faster training
- Lower GPU memory usage
- Faster inference time
- Efficient large-scale retrieval using cosine similarity

Storing processed sequences as `.pk1` files also improves data loading efficiency.

5 Summary

This project successfully developed a deep learning-based gait recognition system for human identification using the CASIA-B dataset.

The system uses silhouette sequence learning with CNN-based feature extraction and Temporal Max Pooling to generate robust gait embeddings. Same-view evaluation achieved excellent performance, while multi-view evaluation demonstrated realistic cross-view challenges.

The model showed strongest performance near side-view angles and lower performance at frontal and rear views, confirming known gait recognition challenges in literature.

The final system is computationally efficient, robust, and suitable for future research improvements such as:

- ResNet backbone
- Triplet Loss
- Horizontal Pyramid Mapping
- GaitSet-style architecture
- Real-time surveillance deployment

6 References

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